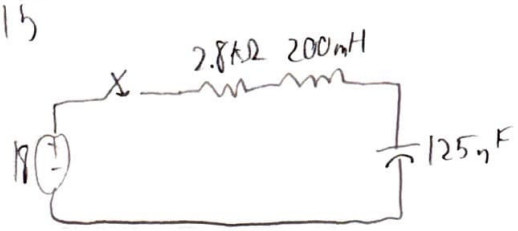
13.15

Dylan Thornburg



$$v_g(s) = \frac{18}{s}$$

$$I(s) = \frac{v_g(s)}{R + sL + \frac{1}{sC}} = \frac{\frac{18}{s}}{2800 + .2s + \frac{8000000}{s}}$$

$$V_C(s) = \frac{1}{sC} I(s)$$

$$V_C(s) = \frac{8000000}{s} \cdot \frac{18}{2800s + .2s^2 + 8000000}$$

$$V_C(s) = \frac{144000000}{2800s^2 + .2s^3 + 8000000s} = \frac{1}{s}\left(\frac{144000000}{.2s^2 + 2800s + 8\times10^6}\right)$$

$$V_C(s) = \frac{1}{s}\left(\frac{7.2\times10^8}{s^2 + 14000s + 4\times10^7}\right) = \frac{A}{s} + \frac{B}{s+10000} + \frac{C}{s+4000}$$

$$\frac{-14000 \pm \sqrt{1.96\times10^8 - 1.6\times10^8}}{2}$$

$$\frac{-14000 \pm 6000}{2} = -10000 \text{ or } -4000$$

$$7.2\times10^8 = A(s+10000)(s+4000) + B(s)(s+4000) + C(s)(s+10000)$$

$s=0$ $\quad 7.2\times10^8 = A(10000)(4000)$

$A = 18$

$s = -10000$ $\quad 7.2\times10^8 = B(-10000)(-6000)$

$B = 12$

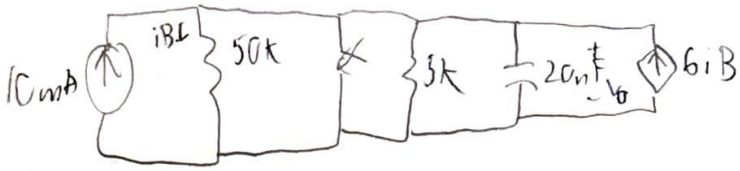
$s = -4000$ $\quad 7.2\times10^8 = C(-4000)(6000)$

$C = -30$

$$V_C(s) = \frac{18}{s} + \frac{12}{s+10000} + \frac{-30}{s+4000}$$

$$V_C(t) = 18 + 12e^{-10000t} - 30e^{-4000t} \text{ V}$$

13.19



$$\frac{-10\times10^{-3}}{s} + \frac{V_o(s)}{50000} + \frac{V_o(s)}{5000} + V_o(s)s(-6\frac{V(s)}{50000}) = 0$$

$$\frac{.01}{s} = \frac{5V_o(s)}{50000} + V_o(s)s(20\times10^{-9})$$

$$\frac{1}{s} = .01V_o(s) + V_o(s)(s)(2\times10^{-6})$$

$$\frac{100}{s} = V_o(s) + V_o(s)(s)(2\times10^{-4})$$

$$\frac{100}{s} = V_o(s)(1+s(2\times10^{-4}))$$

$$\frac{100}{s}\cdot\frac{1}{1+s(2\times10^{-4})} = V_o(s)$$

$$\frac{100}{s(1+s(2\times10^{-4}))} = \frac{A}{s} + \frac{B}{1+s(2\times10^{-4})}$$

$$100 = A(1+s(2\times10^{-4})) + B(s)$$

$$100 = B(-5000)$$

$$B = -\frac{1}{50}$$

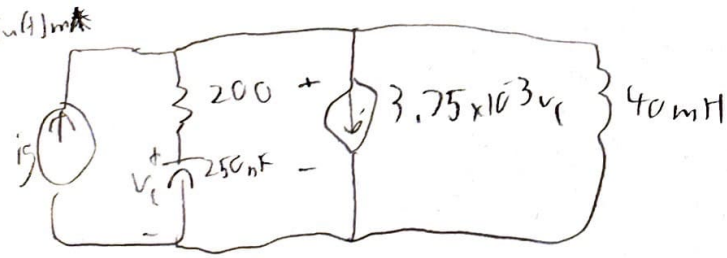
$$\frac{100}{s} = \frac{.02}{1+s(.0002)} = V_o(s)$$

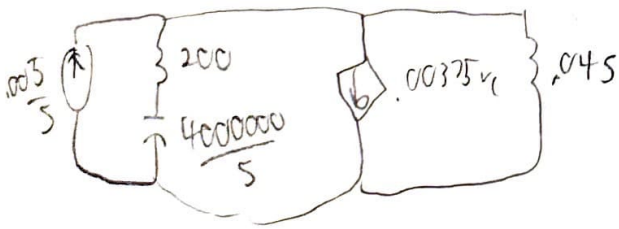
$$\frac{100}{s} - \frac{100}{5000+s} = V_o(s)$$

$$\boxed{100 - 100e^{-5000t} = v_o(t)}$$

13.20

$i_s = 5u(t)$ mA





$$-\frac{.005}{s} + \frac{V_o}{200+\frac{4\times10^6}{s}} + .00375\left(\frac{\frac{4000000}{s}}{200+\frac{4\times10^6}{s}}V_o\right) + \frac{V_o}{.045} = 0$$

$$V_o\left(\frac{1}{200+\frac{4\times10^6}{s}} + \frac{\frac{15000}{s}}{200+\frac{4\times10^6}{s}} + \frac{1}{.045}\right) = \frac{.005}{s}$$

$$V_o\left(\frac{s+15000}{200(s+20000)} + \frac{1}{.045}\right) = \frac{.005}{s}$$

$$V_o\left(\left(\frac{s+15000}{s+20000}\right) + \frac{5000}{s}\right) = \frac{1}{s}$$

$$V_o = \frac{1}{s\left(\frac{s+15000}{s+20000} + \frac{5000}{s}\right)}$$

$$V_o = \frac{s+20000}{s^2+20000s+10^8}$$

$$V_o = \frac{s+20000}{(s+10000)^2} = \frac{A}{s+10000} + \frac{B}{(s+10^4)^2}$$

$$s+20000 = A(s+10000) + B$$

$$B = 10000$$

$$A = 1$$

$B = 10000$

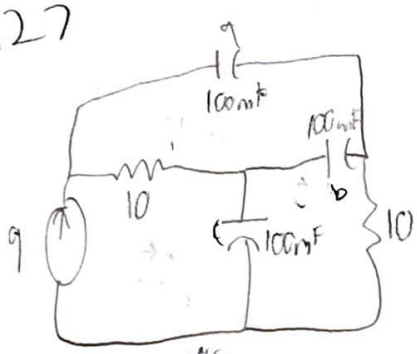
$A = 1$

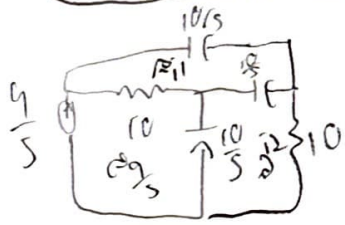
$$V_0 = \frac{1}{s+10000} + \frac{10000}{(s+10000)^2}$$

for $t > 0$;

$$\boxed{V_0(t) = e^{-10000t} + 10000t\,e^{-10000t} \text{ V}}$$

B.27





$$\frac{10}{s} i1 + \frac{10}{s}(i1 - i2) + 10\left(i1 - \frac{9}{s}\right) = 0$$

$$\frac{10}{s}\left(i2 - \frac{9}{s}\right) + 10(i2) + \frac{10}{s}(i2 - i1) = 0$$

$$(s+2)i1 - i2 = 9$$

$$-i1 + (s+2)i2 = \frac{9}{s}$$

$$\begin{vmatrix} (s+2) & -1 \\ -1 & (s+2) \end{vmatrix} \quad (s+2)(s+2) - 1 = s^2 + 4s + 3$$

$$(s+3)(s+1)$$

$$\begin{vmatrix} 9 & -1 \\ \frac{9}{s} & (s+2) \end{vmatrix} = 9s + 18 - \frac{9}{s} \rightarrow \frac{9s^2 + 18s - 9}{s}$$

$$\frac{9}{s}(s^2 + 2s + 1) = \frac{9}{s}(s+1)^2$$

$$i1 = \frac{\frac{9}{s}(s+1)^2}{(s+1)(s+3)} = \frac{9}{s} \cdot \frac{s+1}{s+3}$$

$I_a = i1$ $\quad \frac{9(s+1)}{s(s+3)} = \frac{A}{s} + \frac{B}{s+3}$

$9(s+1) = A(s+3) + B(s)$

$s = -3 \quad -18 = -3B$

$B = 6$

$s = 0 \quad 9 = A3$

$A = 3$

a. $\boxed{I_a = \frac{3}{s} + \frac{6}{s+3}}$

$$\begin{vmatrix} s+2 & 9 \\ -1 & \frac{9}{s} \end{vmatrix} = \frac{9}{s}(s+2) + 9 = 18 + \frac{18}{s} = 18\left(1 + \frac{1}{s}\right)$$

$$= \frac{18}{s}(s+1)$$

$I_b = \frac{9}{s} - i1 \rightarrow \frac{9}{s} - \frac{3}{s} - \frac{6}{s+3} =$

a. $\boxed{I_b = \frac{6}{s} - \frac{6}{s+3}}$

$$\frac{\frac{18}{s}(s+1)}{(s+1)(s+3)} = \frac{18}{s(s+3)} = i2$$

b. $i_a = 3 + 6e^{-3t}$ A

$i_b = 6 - 6e^{-3t}$ A

c. $i_1 = \frac{3}{s} + \frac{6}{s+3} = I_a$, $I_b = \frac{3}{s} - \frac{12}{s+3}$, $I_c = \frac{9}{s} - \frac{6}{s} - \frac{6}{s+3} = \frac{3}{s} - \frac{6}{s+3}$

$i_2 = \frac{18}{s(s+3)} = \frac{6}{s} - \frac{6}{s+3}$

$\frac{18}{s(s+3)} = \frac{A}{s} + \frac{B}{s+3}$

$18 = A(s+3) + Bs$

$s = -3$

$B = -6$

$s = 0$ $A = 6$

$\frac{2}{3} + \frac{3}{4} =$

$V_a = \frac{10}{s}\left(\frac{3}{s} + \frac{6}{s+3}\right)$

$\frac{30}{s^2} + \frac{60}{s^2+3s} = \frac{30(s^2+3s) + 60s^2}{s^4+3s^3} = \frac{90s^2+90s}{s^3(s+3)} = \frac{90s+90}{s^2(s+3)}$

$\frac{90s+90}{s^2(s+3)} = \frac{A}{s^2} + \frac{B}{s} + \frac{C}{s+3}$ $90s+90 = A(s+3) + B(s)(s+3) + Cs^2$

$s=0$ $90 = A(3)$ $A = 30$ $s=-3$ $-180 = 9C$ $C = -20$ $s=1$ $180 = 100 + 4B$ $B = 20$

$$V_a = \frac{30}{s^2} + \frac{20}{s} - \frac{20}{s+3} \text{ V}$$

$V_b = \frac{10}{s}\left(\frac{3}{s} - \frac{12}{s+3}\right) = \frac{30}{s^2} - \frac{40}{s} + \frac{40}{s+3} = V_b$

$V_c = \frac{10}{s}\left(\frac{3}{s} - \frac{6}{s+3}\right) = \frac{30}{s^2} + \frac{20}{s} - \frac{20}{s+3} = V_c$

d.

$V_a(t) = 30t + 20 - 20e^{-3t}$ V

$V_b(t) = 30t - 40 + 40e^{-3t}$ V

$V_c(t) = 30t + 20 - 20e^{-3t}$ V

e. $30t + 20 - 20e^{-3t}$ will reach 1000 volts first

$30t + 20 - 20e^{-3t} = 1000$

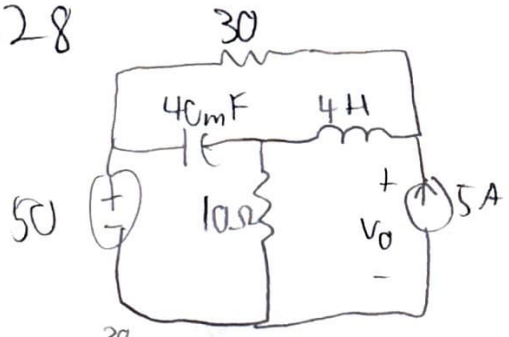
$30t = 980 + 20e^{-3t}$ ← becomes so small we can ignore before we get to 1000

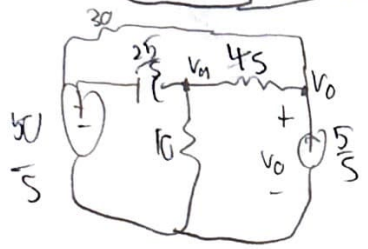
$t = \frac{980}{30} = 32.667$ sec

$30t - 40 = 1000 \rightarrow t = \frac{1040}{30} = 34.667$ sec

when capacitor b breaks

13.28





$$\frac{-V_a}{10} + \frac{V_a - \frac{50}{s}}{\frac{25}{s}} + \frac{V_a - V_o}{4s} = 0$$

$$\frac{V_o - \frac{50}{s}}{30} + \frac{V_o - V_a}{4s} - \frac{5}{s} = 0$$

$$(4s^2 + 10s + 25)V_a - 25V_o = 200s$$

$$\begin{vmatrix} 4s^2+10s+25 & -25 \\ -15 & 2s+15 \end{vmatrix} = 8s(s+5)^2 \qquad (2s+15)V_o + 15V_a = 400$$

$$N_o \begin{vmatrix} 4s^2+10s+25 & 200s \\ -15 & 400 \end{vmatrix} = 200(8s^2+35s+50) \qquad \frac{200(8s^2+35s+50)}{8s(s+5)^2} = \frac{25(8s^2+35s+50)}{s(s+5)^2}$$

$$V_o = \frac{25(8s^2+35s+50)}{s(s+5)^2} = \frac{A}{s} + \frac{B}{s+5} + \frac{C}{(s+5)^2}$$

$$25(8s^2+35s+50) = A(s+5)^2 + B(s)(s+5) + C(s)$$

$s=0$ $A=50$

$s=-5$ $C=-375$

$s=1$ $B=150$

a.

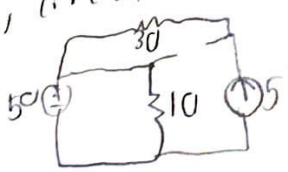
$$V_o = \frac{50}{s} + \frac{150}{s+5} - \frac{375}{(s+5)^2}$$

b.

$$V_o(t) = 50 + 150e^{-5t} - 375te^{-5t}\ V$$

c. at $t=0$

$50+150=200V$; circuit at $t=0$



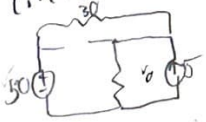
$50 + 30 \cdot 5 = 200$ V

all checks out

at $t=\infty$

$50+0=50V$

circuit at $t=\infty$



$$\frac{V_o(\infty)-50}{3} + \frac{V_o(\infty)}{10} - 5 = 0$$

$$13V_o(\infty) - 650 = 0$$

$$13V_o(\infty) = 650$$

$$V_o(\infty) = 50V$$